Kumar Harsh, M.Sc.

UAV/UAS Systems Test & Certification Engineer | *Robotics, Controls & Automation*

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PROFESSIONAL SUMMARY

Systems engineer specializing in UAV/UAS verification, certification testing, aerospace electrical integration, robotics, and control systems. 5+ years of experience across drone system qualification, battery validation, EMI/EMC, HIL/MIL/SIL testing, flight simulation compliance, smart grid automation, and robotics research. Strong record of converting ambiguous certification needs into test benches, procedures, datasets, verification reports, and regulatory evidence.

CORE COMPETENCIES

Testing & Certification: TEP/ATP documentation, verification reports, conformity evidence, ANATEL, EASA MTQG, FAA, DO-160G, DO-178 methodologies, ASTM, CISPR 22/32

UAV/Robotics Systems: UAV/UAS, VTOL, MALE aircraft, battery systems, BMS, SATCOM, servo firmware, Crazyflie, NAO, RoboMaster, Crazyswarm, ROS/ROS2

Controls & Data: PID, MPC, EKF, SINDYc, system identification, sensor fusion, MATLAB/Simulink, Python, C/C++, R Studio

Tools & Platforms: Git/GitLab, DOORS, SAP, AutoCAD Electrical, Autodesk Inventor, Enterprise Architect, VISIO, Keil, PLC IEC 61131, Raspberry Pi, Arduino

PROFESSIONAL EXPERIENCE

Ground (System) Test Engineer | Wingcopter GmbH, Weiterstadt, Germany **Jun 2024 - Present**

- Developed UAV battery qualification and acceptance test bench setups, enabling repeatable validation and reliability assessment workflows.
- Authored Test Execution Plans (TEP), Acceptance Test Procedures (ATP), verification reports, and compliance documentation for certification activities.
- Executed battery characterization campaigns to determine voltage thresholds and State-of-Charge (SoC) mappings for Immediate Landing and Return-to-Home safety logic.
- Designed a simulated dummy-battery architecture using modified BMS and relay PCBs with external resistor-divider circuitry for destructive qualification testing without damaging production batteries.
- Conducted over-discharge and high-temperature stress testing on simulated battery systems; analyzed multi-UAV operational datasets to define quality thresholds and lifetime test criteria.
- Led UAV weight-envelope statistical analysis using fleet-wide weight and balance reports to support repair conformity assessment.
- Prepared certification test plans, dry runs, conformity evidence, homologation documentation, and regulatory submission packages for ANATEL certification.
- Coordinated conformity taskforce activities across hardware configurations, BOM traceability, PCBA conformity, and certification evidence; led EMI/EMC verification according to CISPR 22/32.
- Conducted servo firmware HIL verification aligned with DO-178 methodologies, lighting qualification per FAA/DO-160G/ASTM standards, and SATCOM latency and robustness testing.

Flight Simulation Engineer | Avengers Flight Group, Dreieich, Germany **May 2024 - Jun 2024**

- Developed EASA-compliant Master Test Quality Guide (MTQG) documentation for Airbus A320/A330 simulator maintenance and qualification.
- Supported simulator compliance activities, maintenance procedures, and technical documentation management for aviation training systems.

Graduate Research Assistant - Smart Grid Automation | IE3, TU Dortmund University, Dortmund, Germany **Dec 2022 - Apr 2024**

- Upgraded the CIMPY 2.4 library to CIMPY 3.0 for IEC 61970-301 standard compatibility.
- Developed an automated framework for generating IID files from ICD templates for IED models in accordance with IEC 61850 standards.
- Built a Python framework for Digital Grid Connection Points and household auto-configuration (.scd) files using CIM data to improve smart-grid control workflows.

Graduate Research Assistant - Human Drone Interaction | FLW, TU Dortmund University, Dortmund, Germany**Mar 2022 - Jan 2024**

- Contributed to a 6DoF IMU dataset for training drone-swarm gesture-control models.
- Improved drone localization through PID tuning and EKF-based sensor fusion integrating IR camera and IMU data, supporting automated charging during swarming operations.
- Supported migration of a ROS-based swarm-show platform to ROS 2 using Crazyswarm for indoor logistics applications; programmed NAO and RoboMaster robots for showcases.
- Enhanced pallet detection by fine-tuning Crazyflie drone controllers for stable imaging with an integrated camera deck.

Executive - Aircraft Design | TATA Advanced Systems, Bangalore, India **Jun 2019 - Aug 2021**

- Developed autopilot control laws and system architecture for Medium Altitude Long Endurance (MALE) aircraft.
- Tested control strategies for robustness in MIL, SIL, and HIL environments; tuned PID controllers to improve VTOL and MALE aircraft stability.
- Designed a precision-focused, laser-powered automated calibration jig.
- Received Star of the Month for prototyping an Invisible UAV using LED screens and pinhole cameras; contributed to i-VTOL UAV team recognized with Best Team Award.

Research Intern | National Institute of Technology, Jamshedpur, India **May 2018 - Jun 2018**

- Modeled and simulated an automatic solar water pump using soil-moisture feedback control.

EDUCATION

Master of Science, Automation and Robotics (Robotics Major) | TU Dortmund University, Germany | Oct 2021 - Feb 2024

- CGPA: 2.1; thesis grade: 1.7. Focus: system identification and controller development.
- Thesis: Nonlinear System Identification and Control using Sparse Identification of Nonlinear Dynamics for Model Predictive Control (SINDYc), evaluated on evoBot with Fraunhofer IML.
- Group project: Upper-body control of NAO soccer robot using MPC for balance during walking and ball kicking, including analytical COM and target-coordinate calculations.

Bachelor of Technology, Electrical and Electronics Engineering | Vellore Institute of Technology, India | Jul 2015 - Apr 2019

- CGPA: 8.67/10. Focus: control systems, smart grids, and power electronics. Thesis: Output Voltage Control of Boost Converter.

PUBLICATIONS

- Kumar Harsh et al., "BLDC Motor Driven Water Pump Fed by Solar Photovoltaic Array using Boost Converter," IJERT, Vol. 09, Issue 06, 2020. DOI: 10.17577/IJERTV9IS060478
- Kumar Harsh et al., "Border Detection System for Fisherman using GPS and Arduino," IJERT, Vol. 08, Issue 12, 2019. DOI: 10.17577/IJERTV8IS120152

CERTIFICATIONS & LANGUAGES

Certifications: Self-Driving Cars Specialization; Python for Everybody; Robotics: Aerial Robotics; MATLAB Comprehensive Training; Aerospace Engineering: Aircraft Fundamentals and Advanced; E3.series for Electrical Engineering.

Languages: English (C1), German (A2), Hindi (C2), Maithili (Native).